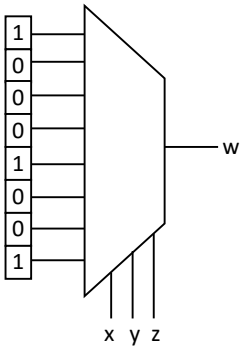
22/11/2017

EE 241 DIGITAL CIRCUITS
MIDTERM

Name:

Q1) (20 points) A digital system is represented as follows.



Find the logic function w=F(x,y,z) in

a- SOP form.
b- POS form.

Q2) (40 points) We have a temperature sensor measuring the values in 10-bit binary form. To convert the read value to Celsius degrees (°C) the following formula should be used.

$$c = (s - 673) * \frac{5}{2}$$

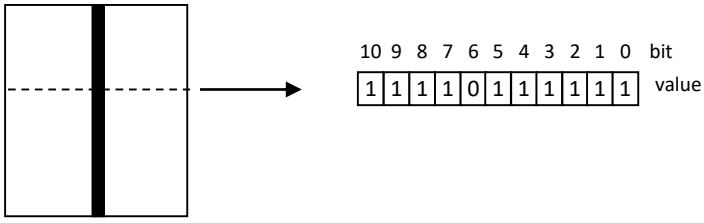
where, s is the sensor measurement and c is the corresponding value in Celsius degrees. The measured temperature can be between 0°C and 63°C.

Design an air conditioner control unit such that, when the measured temperature is less than 16°C, the heater should turn on by feeding it a logic level 1 value. Implement this system in Verilog using

a- Dataflow modeling
b- Behavioral modeling

Q3) (40 points) We will design a path following robot using a line-scan camera, a movement motor, and a direction motor.

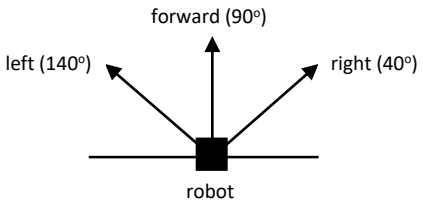
i. The line-scan camera has 11 sensors in array form (c[]). Each sensor gives logic level 1 when a white object is observed. It gives logic level 0 when a black object is observed. The line to be followed will have width one pixel (sensor). As an example,



Black line on white surface

ii. We will not control the movement motor. Hence, we can assume that the robot to be moving in constant speed.

iii. The direction motor needs a 4-bit binary value to set the direction of the robot as follows.



Rotation (degrees)	Motor input
40	5
50	4
60	3
70	2
80	1
90	0
100	-1
110	-2
120	-3
130	-4
140	-5

Motor input will be four bits, negative values will be represented in two's complement form.

Form a robot controller unit with the following specifications.

a- (10 points) We would like to work with inverse input from the line-scan camera. Write a Verilog description to convert bit values from 0 to 1 and from 1 to 0.

b- (10 points) If the robot is moving in straight line, c[5] will be at logic level 1. Hence, the motor input will be 0000.

c- (20 points) If the motor deviates from straight line, then the motor direction should compensate it. As an example, when the camera output is c[10:0]=00000000010 the direction should be 130°. Hence, the motor input will be -4 in binary form.